

RAZA01I2763 -Azacitidine 100 mg (Acc-Mal) PIL
Size: 350 x 350 mm (Final Fold size - 50 x 30 mm)
Color: Pantone Black
Date: 24*04*24

Musculoskeletal, and connective tissue disorders	arthralgia, musculoskeletal pain (includes back, bone and pain in extremity)	Myalgia, muscle spasms,			
Renal and urinary disorders		renal failure", haematuria, elevated serum creatinine	renal tubular acidosis		
General disorders and administration site conditions	pyrexia", fatigue, asthenia, chest pain, injection site erythema, injection site pain, injection site reaction (unspecified)	bruising, haematoma, induration, rash, pruritus, inflammation, discoloration, nodule and haemorrhage (at injection site), malaise, chills, catheter site haemorrhage		injection site necrosis (at injection site)	
Investigations	weight decreased				

- = rarely fatal cases
- Description of selected adverse reactions
- Haematologic adverse reactions
- The most commonly reported adverse reactions associated with azacitidine treatment include anaemia, thrombocytopenia, neutropenia, febrile neutropenia and leukopenia, and were usually Grade 3 or 4. There is a greater risk of these events occurring during the first 2 cycles, after which they occur with less frequency in patients with restoration of haematological function. Most haematological adverse reactions were managed by routine monitoring of complete blood counts and delaying azacitidine administration in the next cycle, prophylactic antibiotics and/or growth factor support (e.g. G-CSF) for neutropenia and transfusions for anaemia or thrombocytopenia as required.
- Infections
- Myelosuppression may lead to neutropenia and an increased risk of infection. Serious adverse reactions such as sepsis, including neutropenic sepsis and pneumonia were reported in patients receiving azacitidine, some with a fatal outcome. Infections may be managed with the use of anti-infectives plus growth factor support (e.g. G-CSF) for neutropenia.
- Bleeding
- Bleeding may occur with patients receiving azacitidine. Serious adverse reactions such as gastrointestinal haemorrhage and intracranial haemorrhage have been reported. Patients should be monitored for signs and symptoms of bleeding, particularly those with pre-existing or treatment-related thrombocytopenia.
- Hypersensitivity
- Serious hypersensitivity reactions have been reported in patients receiving azacitidine. In case of an anaphylactic-like reaction, treatment with azacitidine should be immediately discontinued and appropriate symptomatic treatment initiated.
- Skin and subcutaneous tissue adverse reactions
- The majority of skin and subcutaneous adverse reactions were associated with the injection site. None of these adverse reactions led to discontinuation of azacitidine, or reduction of azacitidine dose in the pivotal studies. The majority of adverse reactions occurred during the first 2 cycles and tended to decrease with subsequent cycles. Subcutaneous adverse reactions such as injection site rash/inflammation/pruritus, rash, erythema and skin lesion may require management with concomitant medicinal products, such as antihistamines, corticosteroids and non-steroidal anti-inflammatory medicinal products (NSAIDs).
- Gastrointestinal adverse reactions
- The most commonly reported gastrointestinal adverse reactions associated with azacitidine treatment included constipation, diarrhoea, nausea and vomiting. These adverse reactions were managed symptomatically with anti-emetics for nausea and vomiting; anti-diarrhoeals for diarrhoea, and laxatives and/or stool softeners for constipation.
- Renal adverse reactions
- Renal abnormalities, ranging from elevated serum creatinine and haematuria to renal tubular acidosis, renal failure and death were reported in patients treated with azacitidine
- Hepatic adverse reactions
- Patients with extensive tumour burden due to metastatic disease have been reported to experience hepatic failure, progressive hepatic coma and death during azacitidine treatment
- Cardiac events
- In patients with newly diagnosed AML, a significant increase in cardiac events is possible with azacitidine treatment.
- Elderly population
- There is limited safety information available with azacitidine in patients ≥ 85 years.
- 4.9 Overdose
- In the event of overdose, the patient should be monitored with appropriate blood counts and should receive supportive treatment, as necessary. There is no known specific antidote for azacitidine overdose.
- 5. Pharmacological properties
- 5.1 Pharmacodynamic properties
- Pharmacotherapeutic group: Antineoplastic agents, pyrimidine analogues; ATC code: L01BC07
- Mechanism of action
- Azacitidine is believed to exert its antineoplastic effects by multiple mechanisms including cytotoxicity on abnormal haematopoietic cells in the bone marrow and hypomethylation of DNA. The cytotoxic effects of azacitidine may result from multiple mechanisms, including inhibition of DNA, RNA and protein synthesis, incorporation into RNA and DNA, and activation of DNA damage pathways. Non-proliferating cells are relatively insensitive to azacitidine. Incorporation of azacitidine into DNA results in the inactivation of DNA methyltransferases, leading to hypomethylation of DNA. DNA hypomethylation of aberrantly methylated genes involved in normal cell cycle regulation, differentiation and death pathways may result in gene re-expression and restoration of cancer-suppressing functions to cancer cells. The relative importance of DNA hypomethylation versus cytotoxicity or other activities of azacitidine to clinical outcomes has not been established.

5.2 Pharmacokinetic properties

- Absorption
- Following subcutaneous administration of a single 75 mg/m² dose, azacitidine was rapidly absorbed with peak plasma concentrations of 750 ± 403 ng/mL occurring at 0.5 h after dosing (the first sampling point). The absolute bioavailability of azacitidine after subcutaneous relative to intravenous administration (single 75 mg/m² doses) was approximately 89% based on area under the curve (AUC).
- Area under the curve and maximum plasma concentration (C_{max}) of subcutaneous administration of azacitidine were approximately proportional within the 25 to 100 mg/m² dose range.
- Distribution
- Following intravenous administration, the mean volume of distribution was 76 ± 26 L, and systemic clearance was 147 ± 47 L/h.
- Biotransformation
- Based on in vitro data, azacitidine metabolism does not appear to be mediated by cytochrome P450 isoenzymes (CYPs), UDP-glucuronosyltransferases (UGTs), sulfoltransferases (SULTs), and glutathione transferases (GSTs).
- Azacitidine undergoes spontaneous hydrolysis and deamination mediated by cytidine deaminase. In human liver S9 fractions, formation of metabolites was independent of NADPH implying that azacitidine metabolism was not mediated by cytochrome P450 isoenzymes. An in vitro study of azacitidine with cultured human hepatocytes indicates that at concentrations of 1.0 µM to 100 µM (i.e. up to approximately 30-fold higher than clinically achievable concentrations), azacitidine does not induce CYP 1A2, 2C19, or 3A4 or 3A5. In studies to assess inhibition of a series of P450 isoenzymes (CYP 1A2, 2B6, 2C8, 2C9, 2C19, 2D6, 2E1 and 3A4) azacitidine up to 100 µM did not produce inhibition. Therefore, CYP enzyme induction or inhibition by azacitidine at clinically achievable plasma concentrations is unlikely.
- Elimination
- Azacitidine is cleared rapidly from plasma with a mean elimination half-life (t½) after subcutaneous administration of 41 ± 8 minutes. No accumulation occurs after subcutaneous administration of 75 mg/m² azacitidine once daily for 7 days. Urinary excretion is the primary route of elimination of azacitidine and/or its metabolites. Following intravenous and subcutaneous administration of ¹⁴C-azacitidine, 85 and 50 % of the administered radioactivity was recovered in urine respectively, while < 1 % was recovered in faeces.
- Special populations
- The effects of hepatic impairment gender, age, or race on the pharmacokinetics of azacitidine have not been formally studied.
- Renal impairment
- Renal impairment has no major effect on the pharmacokinetic exposure of azacitidine after single and multiple subcutaneous administrations. Following subcutaneous administration of a single 75 mg/m² dose, mean exposure values (AUC and C_{max}) from subjects with mild, moderate and severe renal impairment were increased by 11-21%, 15-27%, and 41-66%, respectively, compared to normal renal function subjects. However, exposure was within the same general range of exposures observed for subjects with normal renal function. Azacitidine can be administered to patients with renal impairment without initial dose adjustment provided these patients are monitored for toxicity since azacitidine and/or its metabolites are primarily excreted by the kidney.
- Pharmacogenomics
- The effect of known cytidine deaminase polymorphisms on azacitidine metabolism has not been formally investigated.
- 6. Pharmaceutical particulars
- 6.1 List of excipients
- Mannitol
- 6.2 Incompatibilities
- This medicinal product must not be mixed with other medicinal products.
- 6.3 Shelf life
- 3 years
- After reconstitution:
- When Azacitidine is reconstituted using water for injections that has not been refrigerated, chemical and physical in-use stability of the reconstituted medicinal product has been demonstrated at 25 °C for 45 minutes and at 2 °C to 8 °C for 8 hours.
- The shelf life of the reconstituted medicinal product can be extended by reconstituting with refrigerated (2 °C to 8 °C) water for injections. When Azacitidine is reconstituted using refrigerated (2 °C to 8 °C) water for injections, the chemical and physical in-use stability of the reconstituted medicinal product has been demonstrated at 2 °C to 8 °C for 22 hours.
- From a microbiological point of view, the reconstituted product should be used immediately. If not used immediately, in-use storage times and conditions prior to use are the responsibility of the user and must not be longer than 8 hours at 2 °C to 8 °C when reconstituted using water for injections that has not been refrigerated or not longer than 22 hours when reconstituted using refrigerated (2 °C to 8 °C) water for injections.
- 6.4 Special precautions for storage
- Unopened vials
- Do not store above 30°C.
- Reconstituted suspension
- For storage conditions of the reconstituted of the medicinal product, see section shelf life
- 6.5 Nature and contents of container
- AZADINE (Azacitidine Powder for Suspension for Injection 100 mg/vial) is available in clear type I glass vial sealed with grey siliconised rubber stopper and aluminium flip off white seal. Pack size: Each carton contains 1 glass vial.
- 6.6 Special precautions for disposal and other handling
- Recommendations for safe handling
- Azacitidine is a cytotoxic medicinal product and, as with other potentially toxic compounds, caution should be exercised when handling and preparing azacitidine suspensions. Procedures for proper handling and disposal of anticancer medicinal products should be applied.
- If reconstituted azacitidine comes into contact with the skin, immediately and thoroughly wash with soap and water. If it comes into contact with mucous membranes, flush thoroughly with water.

- Reconstitution procedure
- Azacitidine should be reconstituted with water for injections. The shelf life of the reconstituted medicinal product can be extended by reconstituting with refrigerated (2 °C to 8 °C) water for injections. Details on storage of the reconstituted product are provided below.
- 1. The following supplies should be assembled:
Vial (s) of azacitidine; vial(s) of water for injections; non-sterile surgical gloves; alcohol wipes; 5 mL injection syringe(s) with needle(s).
- 2. 4 mL of water for injections should be drawn into the syringe, making sure to purge any air trapped within the syringe.
- 3. The needle of the syringe containing the 4 mL of water for injections should be inserted through the rubber top of the azacitidine vial followed by injection of the water for injections into the vial.
- 4. Following removal of the syringe and needle, the vial should be vigorously shaken until a uniform cloudy suspension is achieved. After reconstitution each mL of suspension will contain 25 mg of azacitidine (100 mg/4 mL). The reconstituted product is a homogeneous, cloudy suspension, free of agglomerates. The product should be discarded if it contains large particles or agglomerates. Do not filter the suspension after reconstitution since this could remove the active substance. It must be taken into account that filters are present in some adaptors, spikes and closed systems; therefore such systems should not be used for administration of the medicinal product after reconstitution.
- 5. The rubber top should be cleaned and a new syringe with needle inserted into the vial. The vial should then be turned upside down, making sure the needle tip is below the level of the liquid. The plunger should then be pulled back to withdraw the amount of medicinal product required for the proper dose, making sure to purge any air trapped within the syringe. The syringe with needle should then be removed from the vial and the needle disposed of.
- 6. A fresh subcutaneous needle (recommended 25-gauge) should then be firmly attached to the syringe. The needle should not be purged prior to injection, in order to reduce the incidence of local injection site reactions.
- 7. If needed (doses over 100 mg) all the above steps for preparation of the suspension should be repeated. For doses greater than 100 mg (4 mL), the dose should be equally divided into 2 syringes (e.g., dose 150 mg = 6 mL, 2 syringes with 3 mL in each syringe). Due to retention in the vial and needle, it may not be feasible to withdraw all of the suspension from the vial).
- 8. The contents of the dosing syringe must be re-suspended immediately prior to administration. The syringe filled with reconstituted suspension should be allowed up to 30 minutes prior to administration to reach a temperature of approximately 20 °C-25 °C. If the elapsed time is longer than 30 minutes, the suspension should be discarded appropriately, and a new dose prepared. To re-suspend, vigorously roll the syringe between the palms until a uniform, cloudy suspension is achieved. The product should be discarded if it contains large particles or agglomerates.
- Storage of the reconstituted product
- For storage conditions after reconstitution of the medicinal product, see section shelf life
- Calculation of an individual dose
- The total dose, according to the body surface area (BSA) can be calculated as follows:
Total dose (mg) = Dose (mg/m²) x BSA (m²)
- The following table is provided only as an example of how to calculate individual azacitidine doses based on an average BSA value of 1.8 m².

Dose mg/m ² (% of recommended starting dose)	Total dose based on BSA value of 1.8 m ²	Number of vials required	Total volume of reconstituted suspension required
75 mg/m ² (100 %)	135 mg	2 vials	5.4 mL
37.5 mg/m ² (50 %)	67.5 mg	1 vial	2.7 mL
25 mg/m ² (33 %)	45 mg	1 vial	1.8 mL

- Method of administration
- Reconstituted Azacitidine should be injected subcutaneously (insert the needle at a 45-90° angle) using a 25-gauge needle into the upper arm, thigh or abdomen.
- Doses greater than 4 mL should be injected into two separate sites.
- Injection sites should be rotated. New injections should be given at least 2.5 cm from the previous site and never into areas where the site is tender, bruised, red, or hardened.
- Any unused product or waste material should be disposed of in accordance with local requirements.
- 7. Name and Address of Product Registration Holder :
- Accord Healthcare Sdn Bhd (1035160 D)
- Suite 12A-15, Level 12A, Wisma Zelan
- No 1 Jalan Tasik Permaisuri 2
- Bandar Tun Razak
- 56000 Kuala Lumpur
- Manufactured by:
- Intas Pharmaceuticals Limited
- Plot No. 5 to 14, Pharmez Near Village Matoda,
- Tal - Sanand, City: Matoda, Dist.-Ahmedabad,
- Gujarat State, India
- 8. Date of revision of the text
- April 2024
- INP161
- 51 3860 1 730525

(Back Side)

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